Hwarang Kwon

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Research Interests

My research focuses on artificial intelligence (AI), with an emphasis on **Core** (graph representation learning and knowledge-graph-based models) and **Domain** (biomedical and information-centric knowledge graphs). Specifically, I construct and embed large-scale knowledge graphs and leverage these graph representations to mitigate data sparsity and improve downstream tasks such as link prediction, recommendation, and knowledge-intensive question answering.

My key research interests include:
Graph Representation Learning, Knowledge Graph Construction and Embedding, Biomedical Knowledge Graphs, and Knowledge-Graph-based Reasoning and Recommendation.

Education

Myongji University Mar 2025 - Present

- M.S. in Applied Artificial Intelligence
- DataScience Lab / Advisor: Jaehee Jung
- Total GPA of 4.5/4.5

Myongji University Mar 2020 - Feb 2025

- B.S. in Computer Engineering
- Total GPA of 3.49/4.5, Major GPA of 3.66/4.5

Grants

1. **권화랑**, "콜드 스타트 시나리오에서 지식 그래프 표현 학습과 메타러닝을 활용한 약물 단백질 상호작용 예측", *NRF 석사과정생연구장려금지원*, RS-2025-25418034 (₩12,000,000) Sep 2025 - Aug 2026

Publications

Domestic Publications

1. **Effect of Removing the Prediction Target from a Knowledge Graph on Representation Learning and the Classifier**

   Hwarang Kwon, Jaehee Jung

   Korea Software Congress (KSC), 2025

Manuscripts Under Revision

1. **GLANCE: Pair-Level Fusion within Multi-View Representations for Cold-Start Drug–Target Interaction Prediction on Knowledge Graphs**

   Hwarang Kwon, Jeongho Kim, Jaehee Jung

   IEEE Journal of Biomedical and Health Informatics (JBHI) *[Impact Factor: 6.7]*

Research in progress

1. **Region-aware Continual Knowledge Graph Embedding**
2. **Ontology-based Framework for Automatic Construction of a Personalized Knowledge Graph**

Awards & Honors

1. **Best Final Project Award, AI Devcourse**, *Programmers* 2023

Extracurricular Activities

1. **Member**, AI Paper Review Study Group May 2023 - Aug 2023
2. **Member**, AI Competition Study Group May 2023 - Aug 2023
3. **Trainee**, AI Devcourse, Programmers Mar 2023 - Aug 2023
4. **Trainee**, Advanced AI Course, Daegu AI Hub Aug 2022 - Oct 2022

Internships

DataScience Lab, Advisor: Jaehee Jung October 2024 - Feb 2025

- Undergraduate Researcher
- Participated in paper seminars

Skills

English: OPic: IM1

Programming: Python, C

Deep Learning: Pytorch, Tensorflow, Scikit-learn, Keras

Data Analysis: Numpy, Pandas, Matplotlib, Seaborn